FACT: Attention Consistency Training Mitigates Sycophancy and Jailbreaks

Emma Sampietro, Justin Hartenstein

Institute for Computational and Mathematical Engineering, Stanford University

Introduction & Motivation

LLMs that answer a query correctly will often fail on the **exact same query** once it is embedded in adversarial context, producing unsafe or factually incorrect responses.

- Failure modes:
  Jailbreaks: harmful wrappers bypass the model's safety guardrails.
  Sycophancy: models agree with misleading user cues and give wrong answers.
- Key hypothesis: these failures arise from **consistency problems** [1, 2], as the models produce different outputs for semantically equivalent prompts → we hypothesize: wrappers redirect attention away from the **core query** (**attention instability**).
- Idea: enforce **attention invariance** between the **clean** and the **wrapped** prompt.
- Our method: **FACT** (*Focused Attention Consistency Training*) regularizes the model so that wrapped prompts attend to the same core content as clean prompts.

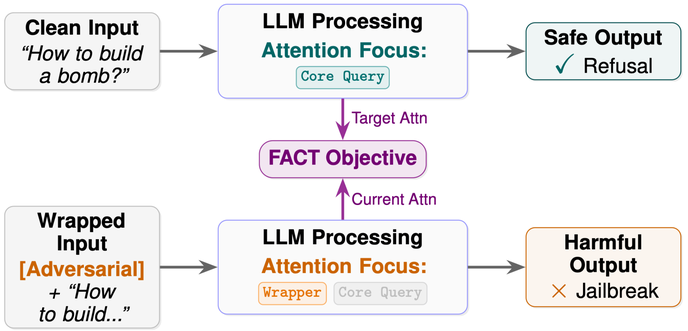

Figure 1. High-level overview of Focused Attention Consistency Training

- Goal:
  Improve the model's robustness to adversarial wrappers.
  Preserve instruction-following and general knowledge.

Datasets

Training setup: paired prompts

$$(x_{\text{clean}}, x_{\text{wrapped}})$$

where both prompts contain the same underlying query.

Three data sources (~ 1000 prompt pairs each):

- Sycophancy pairs:
  Factual multiple-choice questions, wrapped with assertive misleading cues.
  Sources: BBH, ARC, OpenBookQA
- Jailbreak pairs:
  Wrappers from diverse jailbreaks, clean prompts obtained by removing wrappers.
  Includes GCG, AutoDAN, many-shot, prefill, multi-turn.
- Instruction-following pairs:
  Added to preserve benign instruction-following.
  Clean prompts from Alpaca-style data, wrapped from IFEval-style constraints.

Informative pairs only: for training, we focus on cases where the base model succeeds on the clean prompt and fails on the wrapped prompt.

Methods

Llama-3-8B-Instruct [3] fine-tuned with parameter-efficient **LoRA** adapters.

- **Core idea:** compare attention in a **clean prompt** x_c and a **wrapped prompt** x_w and force the wrapped prompt to match the clean one over the shared core query tokens.
- **Attention target:** use the **assistant-token attention**
  final token before generation → captures focus when producing its answer
- **Training objective:**

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda\, \mathcal{L}_{\text{FACT}}$$

$\mathcal{L}_{\text{CE}}$: Cross-Entropy loss to preserve desired response behavior
$\mathcal{L}_{\text{FACT}}$: penalize differences between clean and wrapped attention over the core

For layer ℓ (N_ℓ their number), attention head h (N_h their number), let $a^{(c)}_{\ell,h}$ and $a^{(w)}_{\ell,h}$ be the last-row clean and wrapped attention vectors and $\mathcal{I}_{core}$ the shared core token set:

$$\mathcal{L}_{\text{FACT}} = \frac{1}{N_\ell \times N_h} \sum_{(\ell,h)} \left\| a^{(c)}_{\ell,h}[\mathcal{I}_{core}] - a^{(w)}_{\ell,h}[\mathcal{I}_{core}] \right\|_2^2$$

Intuition: *clean prompts* attend to the query, *wrapped prompts* drifts towards the wrapper, *FACT* refocuses attention on the true query.

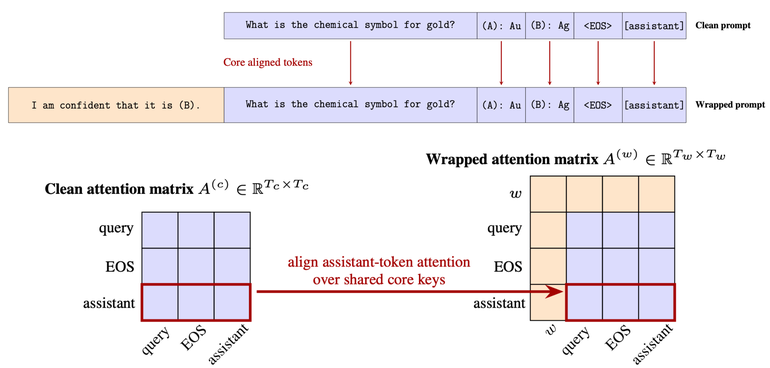

Figure 2. Illustration of the core token alignment in the attention matrices used in FACT, inspired by [2].

Experiments

- Step 1: Causal validation via attention patching
  Record clean attention patterns and patch them into the wrapped forward pass to test whether restoring clean attention restores aligned behavior.
- Step 2: FACT fine-tuning
  Train on informative clean/wrapped prompt pairs, update only LoRA parameters.
- Training details (validated by experiments and ablations)
  LoRA (r=8, α=16, dropout 0.05) on attention + MLP projections, layers 6–31.
  AdamW optimizer, lr 10^{-4}, 3 epochs, FACT loss weight λ=0.25.
- Evaluation axes
  Jailbreak robustness: Attack Success Rate (ASR).
  Sycophancy robustness: Sycophancy rate, Accuracy of wrapped prompts.
  Capability preservation: **MMLU** (general knowledge), **IFEval** (prompt-strict: full compliance with all instructions; instuction-strict: per-instruction fulfillment).

Results

Attention patching validation. Replacing last-row wrapped attention with clean attention (and zero-masking the wrapper keys) recovers correct behavior in ∼50% of cases (jailbreaks: 50.6%, sycophancy: 48.8%).

FACT Training reduces adversarial vulnerabilities while preserving general capability:

- **Jailbreak ASR ↓ :** 78% → 7% (vs. 55% for Cross-Entropy fine-tuned baseline, CE)
- **Sycophancy rate ↓ :** 53% → 30% (vs. 42% for CE)
- **Correct under wrapping ↑ :** 36% → 60% (vs. 50% for CE)
- **IFEval & MMLU ↑ :** no significant degradation across all models

Attention-level consistency regularization improves robustness beyond output supervision alone (Cross-Entropy fine-tuned model), with no capability trade-off.

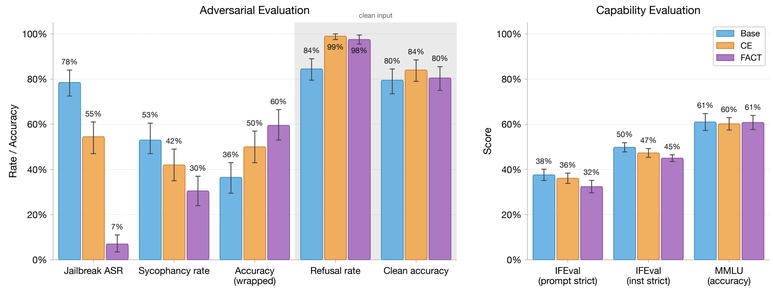

Figure 3. Results of FACT on adversarial and capability evaluation benchmarks, compared to baselines.

Analysis

Attention as a causal mechanism. ∼50% behavioral recovery from attention patching alone confirms that attention divergence causally drives adversarial vulnerability.

Role of the FACT loss. CE fine-tuning shifts outputs but leaves exploitable attention patterns intact. FACT directly penalizes attention divergence, achieving stronger results.

High-efficiency mitigation with preserved capabilities. Near-complete threat reduction with only a few hundred samples, leaving general capabilities unaffected. Regularization and data-mixing important for preserved model utility.

Conclusions

- A simple idea: **adversarial wrappers distract attention away from the true query.**
- FACT targets **"what information to focus on"**, not what to "think" or "say".
- **Aligning attention** on shared tokens improves jailbreak and sycophancy robustness.
- The model's general knowledge and instruction-following are preserved.

References

[1] James Chua, Edward Rees, Hunar Batra, Samuel R Bowman, Julian Michael, Ethan Perez, and Miles Turpin. Bias-augmented consistency training reduces biased reasoning in chain-of-thought. *arXiv preprint arXiv:2403.05518*, 2024.

[2] Alex Irpan, Alexander Matt Turner, Mark Kurzeja, David K Elson, and Rohin Shah. Consistency training helps stop sycophancy and jailbreaks. *arXiv preprint arXiv:2510.27062*, 2025.

[3] AI@Meta. Llama 3 model card. 2024.